

Experiment 7: Depth First Search (DFS)

Aim

To implement the Depth First Search (DFS) algorithm using Python.

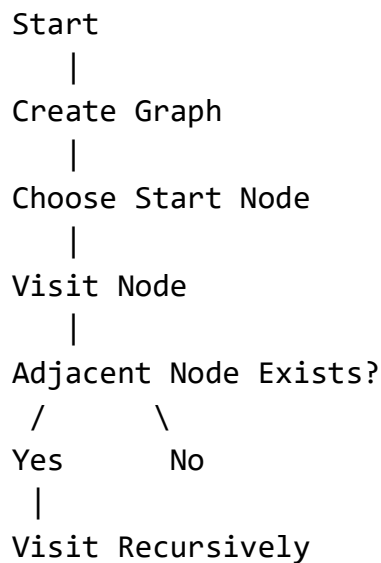
Objective

- To traverse all nodes using recursion.
- To explore graph nodes deeply before backtracking.

Algorithm

1. Create a graph.
2. Select the starting node.
3. Mark it as visited.
4. Visit each adjacent unvisited node recursively.
5. Continue until all nodes are visited.

Flowchart



```
|  
Repeat  
|  
End
```

Program

```
graph = {  
    'A': ['B', 'C'],  
    'B': ['D', 'E'],  
    'C': ['F'],  
    'D': [],  
    'E': ['F'],  
    'F': []  
}  
  
visited = set()  
  
def dfs(node):  
    if node not in visited:  
        print(node, end=" ")  
        visited.add(node)  
        for i in graph[node]:  
            dfs(i)  
  
dfs('A')
```

Output

A B D E F C

Result

The DFS traversal was executed successfully.

Conclusion

DFS explores graph nodes deeply using recursion or a stack.

Result

The BFS traversal was performed successfully.

Conclusion

The Breadth First Search algorithm visits nodes level by level using a queue.